Rates of an acid-catalysed hydrolysis

A. Student

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Introduction

Ester hydrolysis in dilute acid follows pseudo-first-order kinetics when water is in large excess (Laidler, 1987). The rate law is

$$-\frac{d[\mathrm{E}]}{dt} = k_{\mathrm{obs}}\,[\mathrm{E}], \qquad k_{\mathrm{obs}} = k\,[\mathrm{H^+}]$$

and the integrated form $\ln[\mathrm{E}]_t = \ln[\mathrm{E}]_0 - k_{\mathrm{obs}} t$ gives a straight line.

Method

Samples were titrated at fixed intervals with 0.100 M NaOH. Each run was repeated three times.

Reagents

- ethyl acetate, 0.50 M
- hydrochloric acid, 1.0 M
- sodium hydroxide, 0.100 M (standardised)

Procedure

1. Mix the ester and the acid in a thermostatted flask.
2. Withdraw 5.00 mL every five minutes.
3. Quench in ice and titrate.

Results

t / min	V(NaOH) / mL	ln(V∞ − Vt)
0	12.1	2.75
10	14.3	2.51
20	16.0	2.27
30	17.4	2.01

The slope of the fitted line is $-0.0247\ \mathrm{min^{-1}}$, so $k_{\mathrm{obs}} = 4.1 \times 10^{-4}\ \mathrm{s^{-1}}$.

Discussion

The value agrees with the literature within 8 % (Laidler, 1987). The largest error is the quench time; see `notes.md`.

References

Laidler, K. J. (1987). *Chemical kinetics* (3rd ed.). Harper & Row.